Chirag Garg

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.30
B.S.M Public School	Class XII (CBSE)	2023	93.40%
New Lahoria Vidya Mandir Sr. Sec. School	Class X (CBSE)	2021	95.00%

WORK EXPERIENCE

Hardware Engineering Intern, Ceremorphic

- Validated PCIe Gen6 receiver architecture (64 Gbps/lane) through Digital Mixed-Signal (DMS) testing, integrating digital and behavioral analog models to ensure proper loop convergence and functional signoff.
- Evaluated multi-state sequencing FSMs for critical receiver calibration and validated digital offset correction logic using Exponential Moving Averages (EMA) to eliminate offsets across CTLE stages and PAM4 slicers.

PROJECTS

Mixed-Signal IC Design: Successive Approximation Register (SAR) ADC

Guide: Dr. Nitesh Singhal, Dr. Gajendranath Chaudhury

- Designed and simulated a 10-bit SAR ADC using TSMC 180nm CMOS technology in Cadence Virtuoso.
- Implemented blocks like capacitive DAC, strong-arm comparator, bootstrapped switch and SAR logic.
- Conducted comprehensive mixed-signal simulations (Transient, Noise, Monte Carlo) via Cadence ADE.

RISC-V 32-bit Processor

- Implemented the microarchitecture of a 5-stage pipelined processor in Verilog for the RV32I ISA.
- Optimized performance by hazard resolution techniques like stalling, flushing, and data forwarding.
- Verified the funcitoning by using a python based assembler to feed instructions to internal memory.

SDRAM Controller Design in Verilog

- Designed and verified an SDRAM controller in Verilog HDL with modules for initialization, auto-refresh, selfrefresh, mode register programming, and read/write operations using FSM-based control.
- Simulated and tested functionality with Micron SDRAM behavioral model, gaining hands-on experience in memory subsystem design and DRAM timing.

AES-128 Cryptographic Processor

- Designed and implemented an AES-128 encryption/decryption module in Verilog HDL, supporting 128-bit plaintext, ciphertext, and key processing.
- Implemented core AES hardware operations including SubBytes, ShiftRows, MixColumns, AddRoundKey, and 128-bit Key Expansion using synthesizable RTL.
- Verified the RTL design using a Verilog testbench and standard AES test vectors, ensuring functional correctness of encryption and decryption across all 10 rounds.

VLSI Design Flow

- Designed and implemented a Booth multiplier, executing the complete VLSI flow using open-source tools.
- Performed lint checking for design correctness using Verilator, and synthesized the design using Yosys.
- Executed the backend flow using OpenROAD via Tcl scripting and closed timing using OpenSTA.

SKILLS

Tools: Xilinx Vivado, Cadence Virtuoso, KiCad, LTSpice, VS Code, Linux, GTKWave

Languages: Verilog, C, Assembly (RISC-V), Python, LaTeX

PD Toolchain: Yosys, OpenSTA, Openroad

POSITIONS OF RESPONSIBILITY

Core Member, Elektronica (Electronics Club)

- Worked on the construction of an 8-bit computer on breadboards. Designed core components like ALU, registers, clock, and memory using discrete ICs, flip-flops, and logic gates.

Teaching Assistant, Math for EE (EE1010)

- Mentored 50+ students through doubt sessions and exam feedback to strengthen conceptual understanding.

Placement Coordinator, IIT Hyderabad

- Pitched companies for IIT Hyderabad placements and managed smooth OCS portal registration.

RELEVANT COURSES

Digital: Digital Systems, Digital Systems Lab (Verilog), Computer Architecture, Digital IC Design

Analog: Analog Electronics, Analog Lab, Analog IC Design, Semiconductor Device Fundamentals, Control Theory

Fundamental Courses: Circuits and Network Analysis, Introduction to Programming(C), Signals and System

EXTRACURRICULAR

Represented IIT Hyderabad in the Inter-IIT Cultural Meet 6.0 at IIT Kharagpur as part of the dramatics team.

Contributed actively as a volunteer with the National Service Scheme (NSS).